

Surface Movement Guidance and Control System (SMGCS)

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(SMGCS)

- **To enhance taxiing capabilities in low visibility with additional markings and lights.**

- **Required for airports which have Air Carrier takeoff or landing operations with less than 1200 feet runway visual range (RVR).**

- **Requires airports to have a low visibility taxi plan.**

- **Affects both aircrew and vehicle operators.**

Surface Movement Guidance and Control System (SMGCS)

Special requirements are in effect for qualified airline crews when operating below 1200 RVR. ***(Does not apply to Part 91 operators)***

Special markings and lighting are installed at airports to support SMGCS operations. Airports conducting SMGCS operations must also have a published SMGCS Plan.

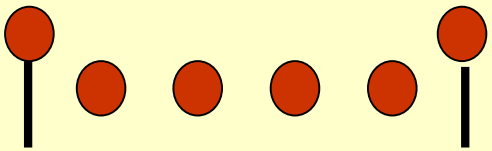
Surface Movement Guidance and Control System (SMGCS)

Two Levels:

- **Below 1200 RVR down to and including 600 RVR**

- **Below 600 RVR**

SMGCS Marking and Lighting Features



Stop Bar Lights

Row of red, **in-pavement** lights that when illuminated designate a runway hold position. **Elevated** stop bar lights are additional features at those airports where surface conditions (snow) may cover the in-pavement lights.



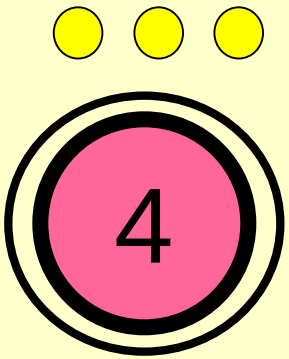
Runway Guard Lights

Elevated or in-pavement yellow flashing lights installed at runway holding positions.



Taxiway Centerline Lights

Green in-pavement lights to assist taxiing aircraft in darkness and in low visibility conditions.



Clearance Bar Lights

Geographic Position Marking

In-pavement yellow lights, when installed with geographic position marking “pink spot”, indicate the designated aircraft or vehicle hold point location on the airport surface.

Pilot Visual Aids for SMGCS



Taxiway centerlines on SMGCS routes are 12” wide and highlighted in black.

Pilot Visual Aids for SMGCS

Elevated runway guard lights “Wig Wags” are located on both sides of the hold position markings for SMGCS runways. If the runway intersection is over 150 feet wide, runway guard lights are installed in-pavement in front of the hold position marking at 10 foot intervals. These flashing yellow lights are used to enhance the identification of runway hold positions during low visibility operations. Runway guard lights may also be used at “high alert” intersections at other airports..





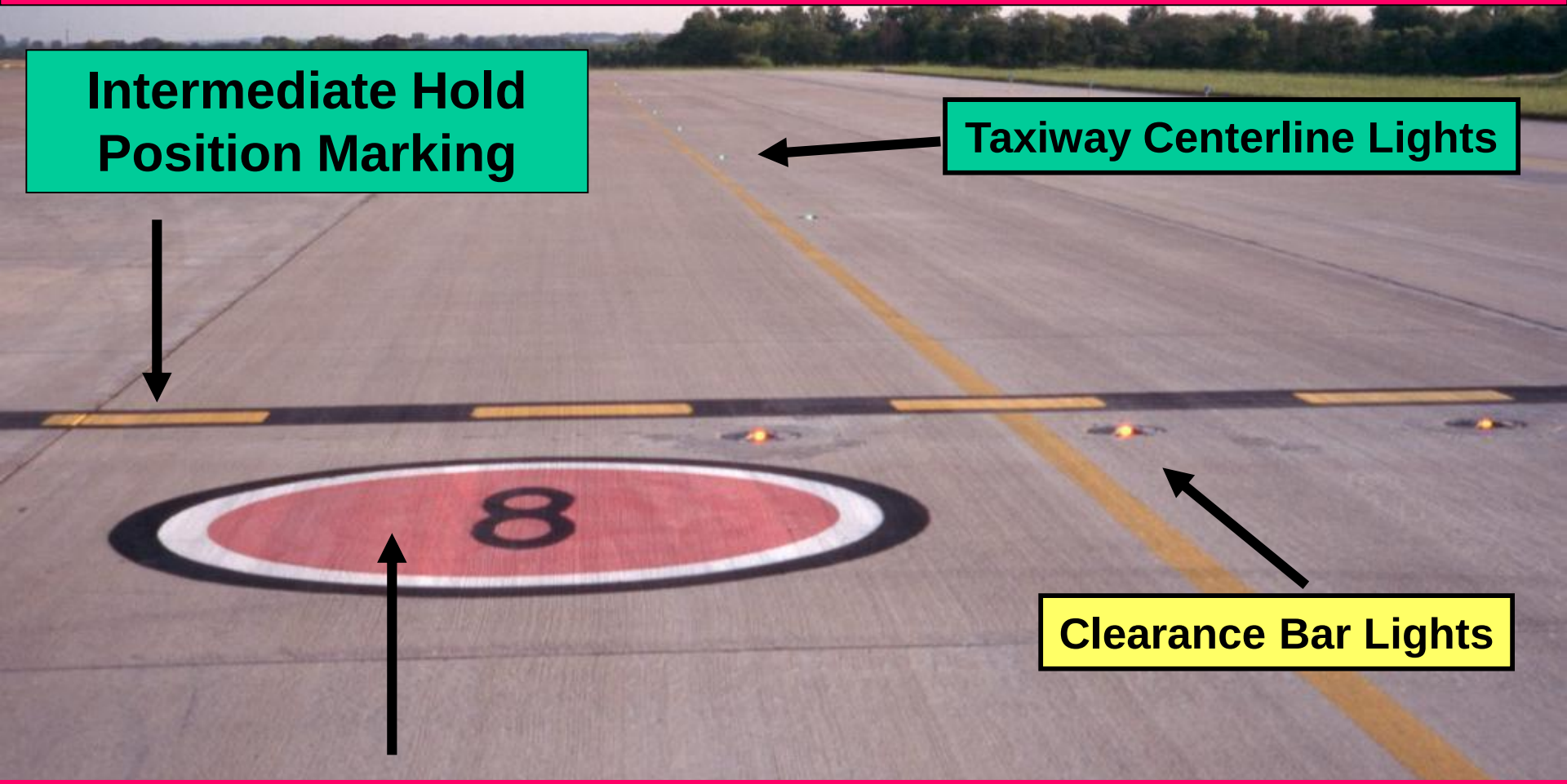
Elevated Runway Guard Lights - “Wig Wags”

Pilot Visual Aids for SMGCS



Runway Guard Lights are installed at all hold positions for SMGCS runways.

Additional SMGCS Pilot Visual Aids at Airports Approved to Operate BELOW 600 RVR



Geographic Position Markings (“pink spots”) are used at taxiway hold points for staging aircraft during SMGCS operations.

Additional SMGCS Pilot Visual Aids at Airports Approved to Operate BELOW 600 RVR

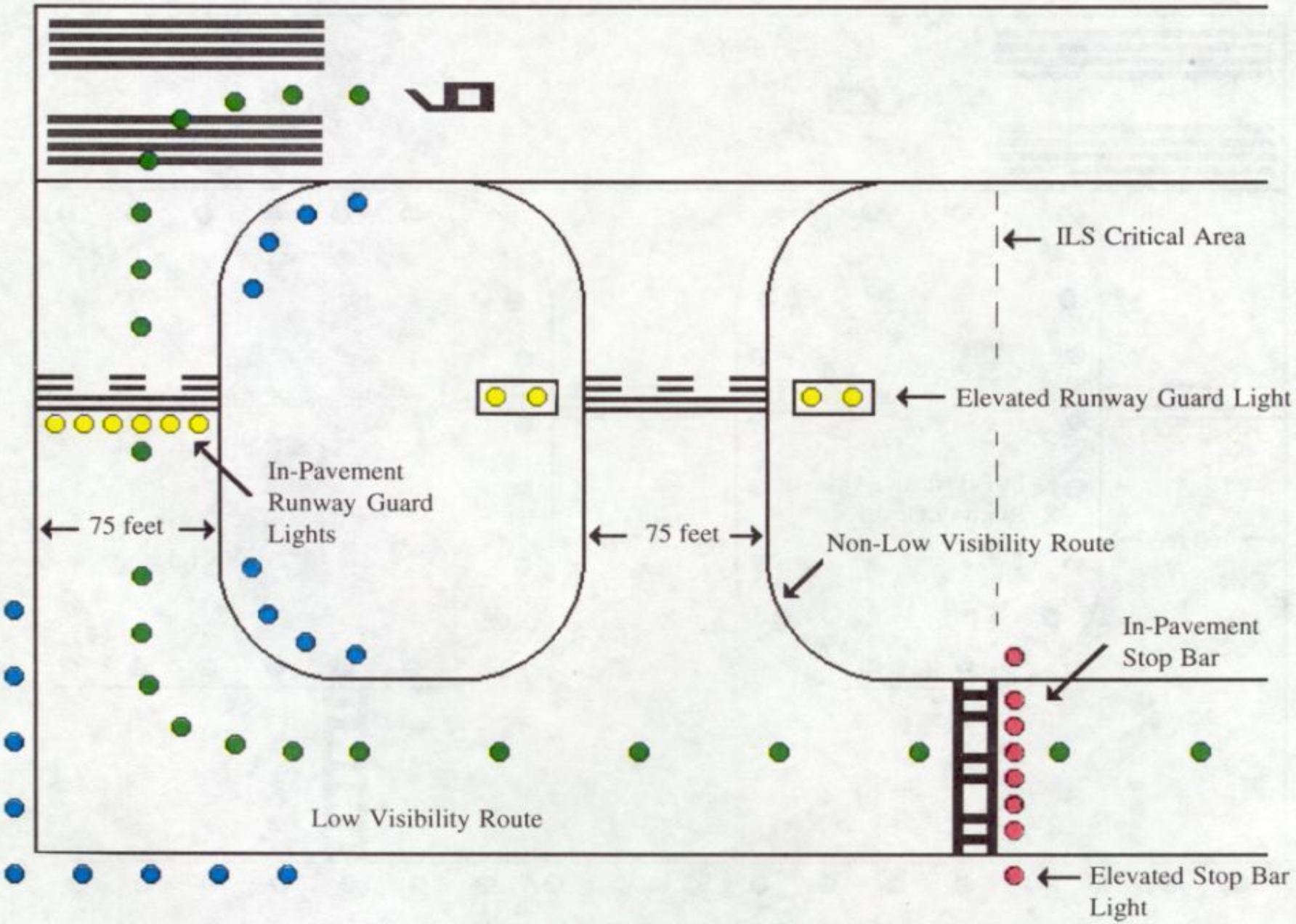


To support airline operations below 600 RVR, Clearance Bars are installed at hold points on taxiways. Taxiway centerline lights are installed on all SMGCS routes.

Additional SMGCS Pilot Visual Aids at Airports Approved to Operate BELOW 600 RVR

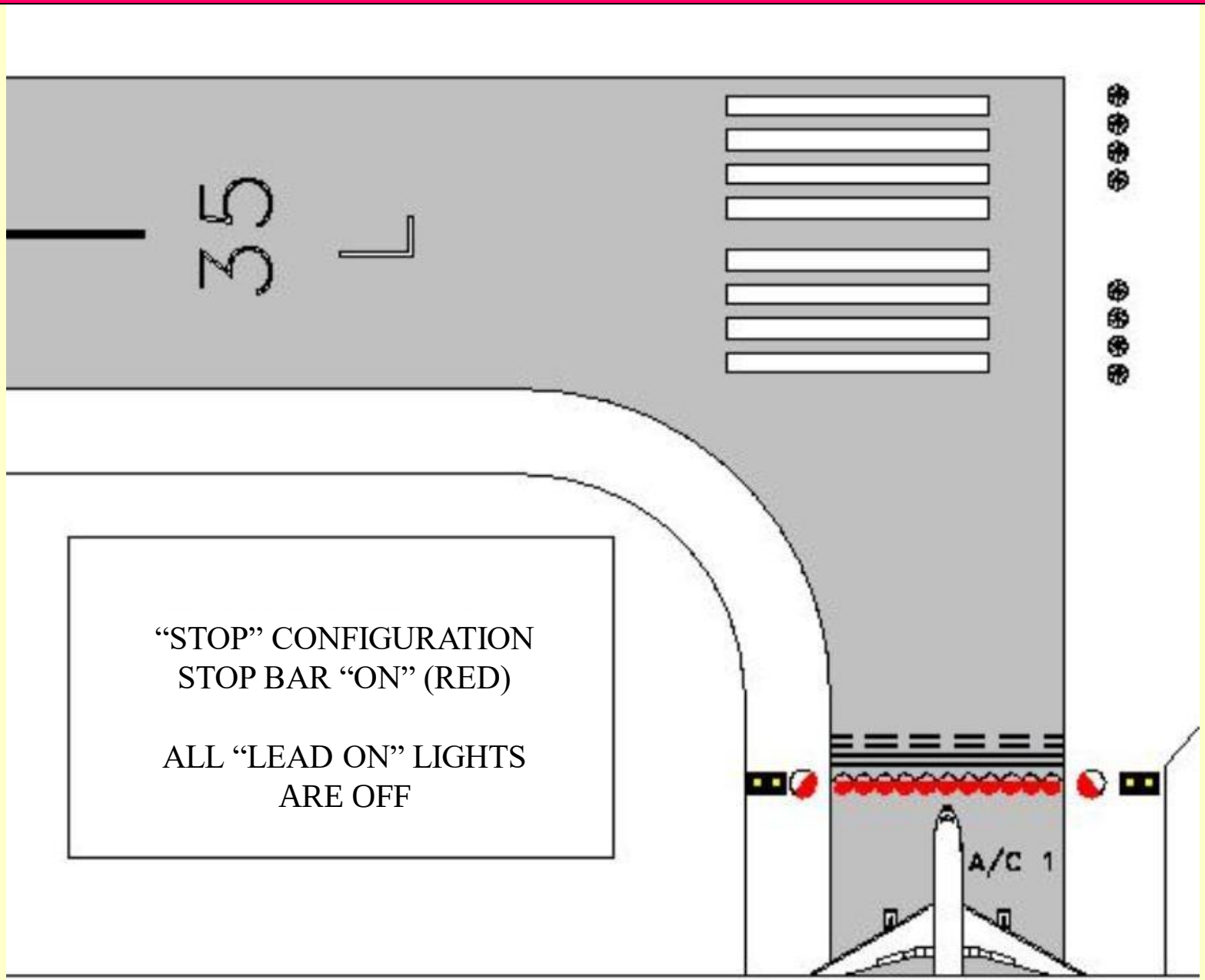


Stop Bar light fixtures are located across the taxiway pavement and are 9 feet 10 inches apart.

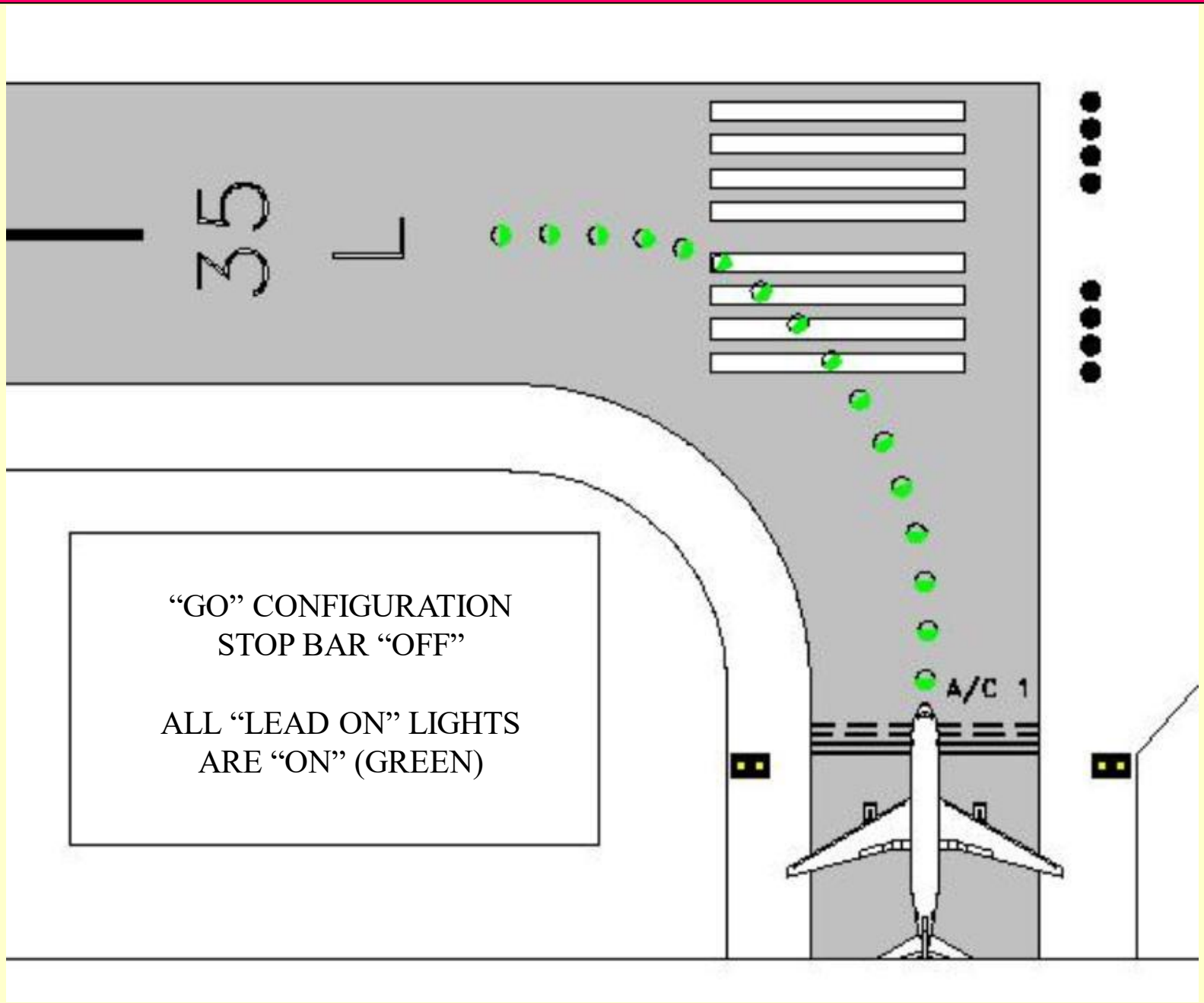


Example of SMGCS Lighting-Below 600 RVR

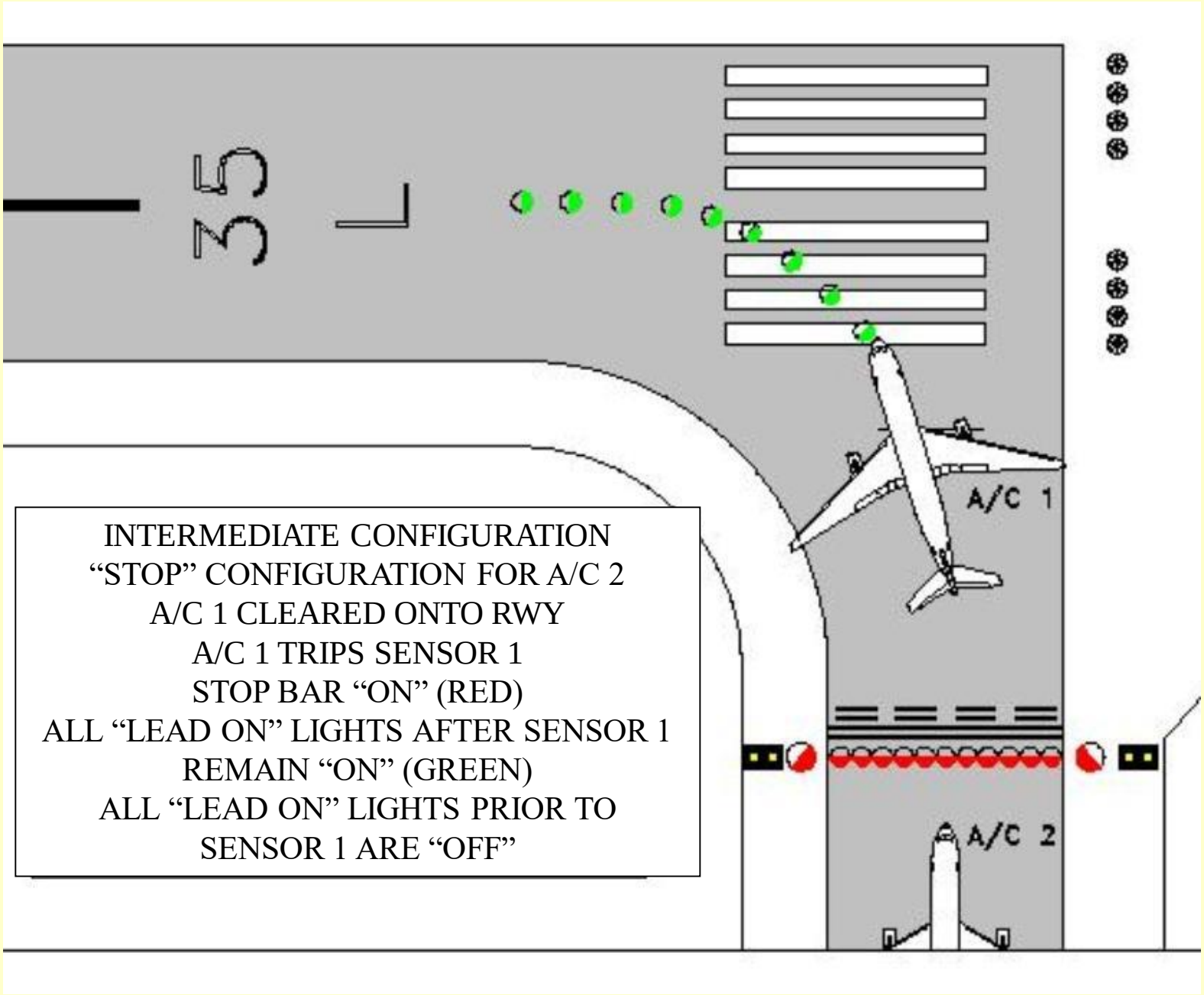
Pilot Visual Aids for SMGCS-Below 600 RVR



Pilot Visual Aids for SMGCS-Below 600 RVR

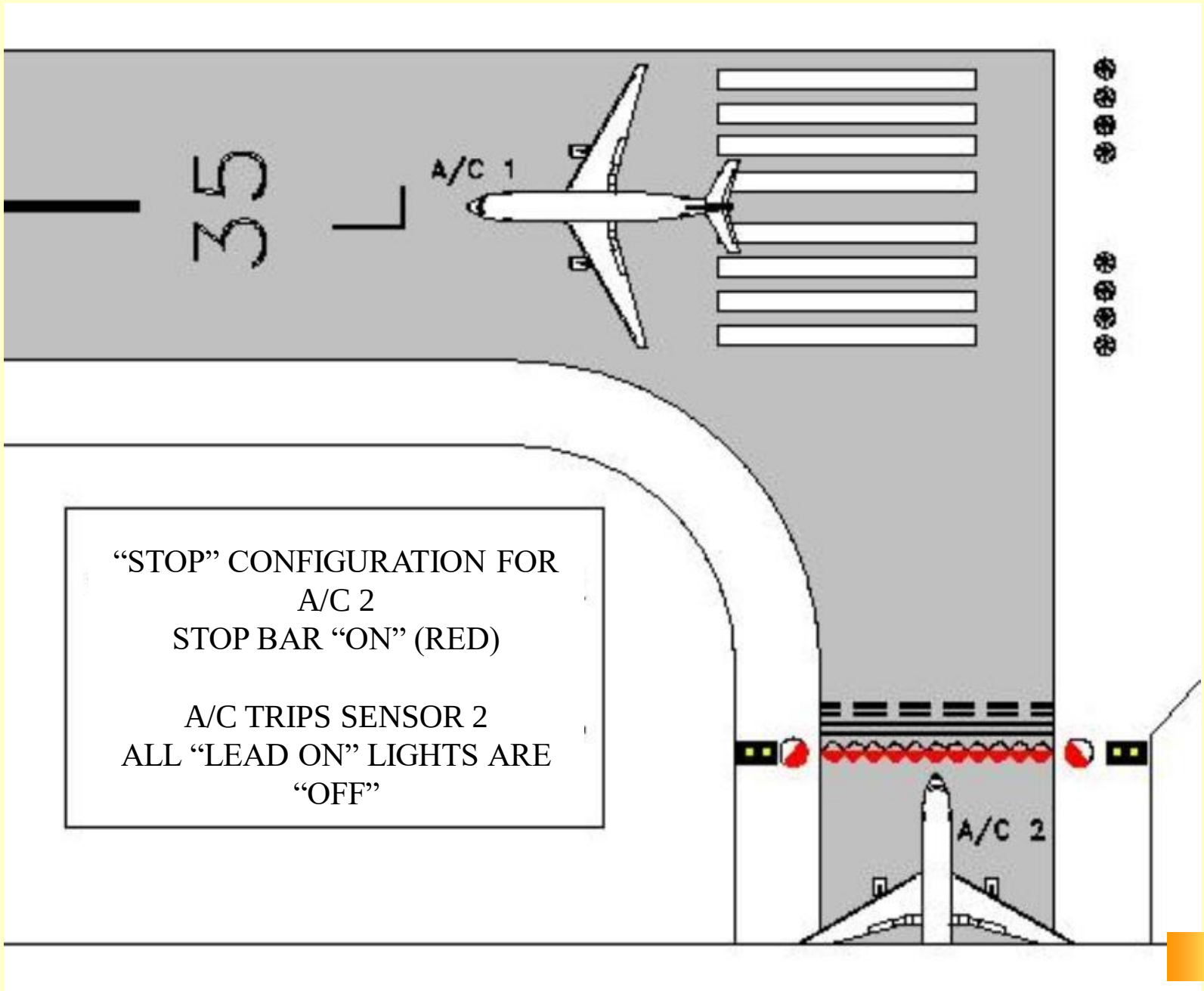


Pilot Visual Aids for SMGCS-Below 600 RVR



INTERMEDIATE CONFIGURATION
"STOP" CONFIGURATION FOR A/C 2
A/C 1 CLEARED ONTO RWY
A/C 1 TRIPS SENSOR 1
STOP BAR "ON" (RED)
ALL "LEAD ON" LIGHTS AFTER SENSOR 1
REMAIN "ON" (GREEN)
ALL "LEAD ON" LIGHTS PRIOR TO
SENSOR 1 ARE "OFF"

Pilot Visual Aids for SMGCS-Below 600 RVR



Exit